

Yeast Fermentation and Fuel Sources

Student worksheet | Biology 2e, Chapter 7 | 45-60 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

How does the available sugar affect gas accumulation in a yeast culture?

Compare baker's yeast with no added sugar, glucose, and sucrose. Record cumulative gas volume and compare average accumulation rates over the same time interval.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: 0.5 g baker's yeast, 50 mL final mixture, 30 deg C water bath, identical gas collection apparatus, 30-minute observation. Sugar treatments have equal mass concentration, not equal molarity.

Select your route

Part	Virtual / Classroom	Equipment or conditions
No added sugar		
Glucose		
Sucrose		

Use the same questions for either route. Always label the source of each dataset.

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2. Collect observations

Record collected gas (ml). Repeat each treatment with fresh material for three independent trials.

Treatment / trial	0 min	5 min	10 min	15 min	20 min	25 min	30 min
No added sugar / 1							
No added sugar / 2							
No added sugar / 3							
Glucose / 1							
Glucose / 2							
Glucose / 3							
Sucrose / 1							
Sucrose / 2							
Sucrose / 3							

Source (virtual/classroom) for each treatment: _____

Trial summaries

Treatment	Trial 1	Trial 2	Trial 3	Mean
No added sugar				
Glucose				
Sucrose				

Metric: Average gas accumulation rate in mL/min = (volume at 30 min - volume at 0 min) / 30.

Observations and graph

Graph time on the x-axis and the recorded measurement on the y-axis on separate graph paper, or use the app. Label treatments, units, and data sources. Record unusual observations below.

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3. Explain the evidence

Average gas accumulation rate = (final volume - initial volume) / elapsed minutes. Gas from yeast is consistent with CO₂ production, but volume alone does not prove fermentation rather than aerobic metabolism or directly measure ATP yield.

1. Compare the mean gas accumulation rates for the three treatments. Use numerical evidence and discuss overlap between trials.

2. Why include yeast with no added sugar? What additional control would help separate yeast activity from physical gas expansion?

3. Explain how fermentation supports continued glycolysis. Why can gas volume alone not establish the amount of ATP made or prove oxygen was absent?

4. Identify one limitation of equal-mass sugar comparisons or gas collection, and propose a better experiment.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.